

DYNAMIC PRICE OPTIMISATION FOR URBANMART

Evidence-based SKU pricing for Jan-Mar 2025

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Agenda

- Project goal, data and analytical approach.



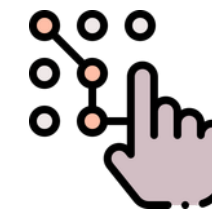
- Insight 1: Where are the current margin opportunities?



- Insight 2: How is UrbanCart positioned against competitors?



- Insight 3: Which categories are most price-sensitive?



- Price recommendation and optimization model.



- Q1 2025 simulated business impact.



- Implementation roadmap and next steps.



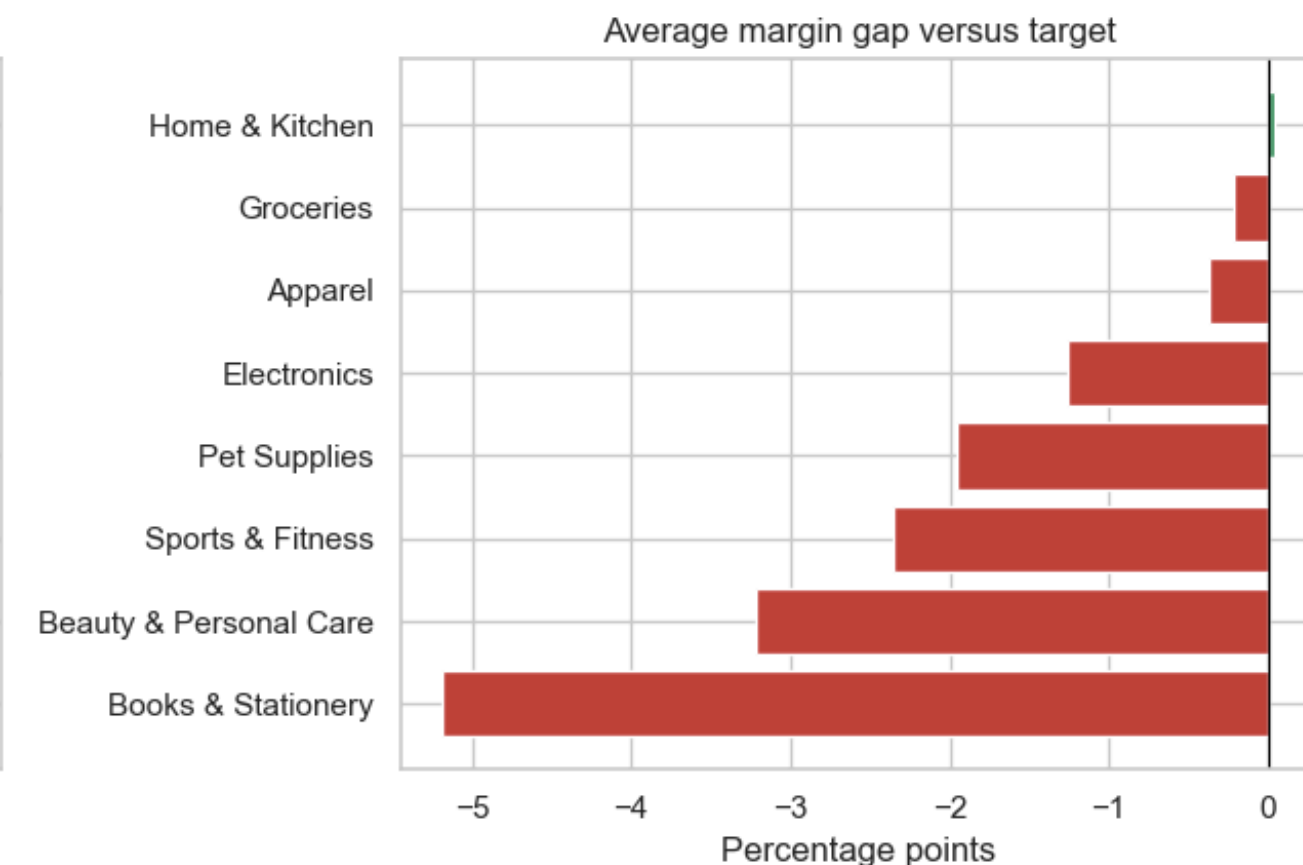
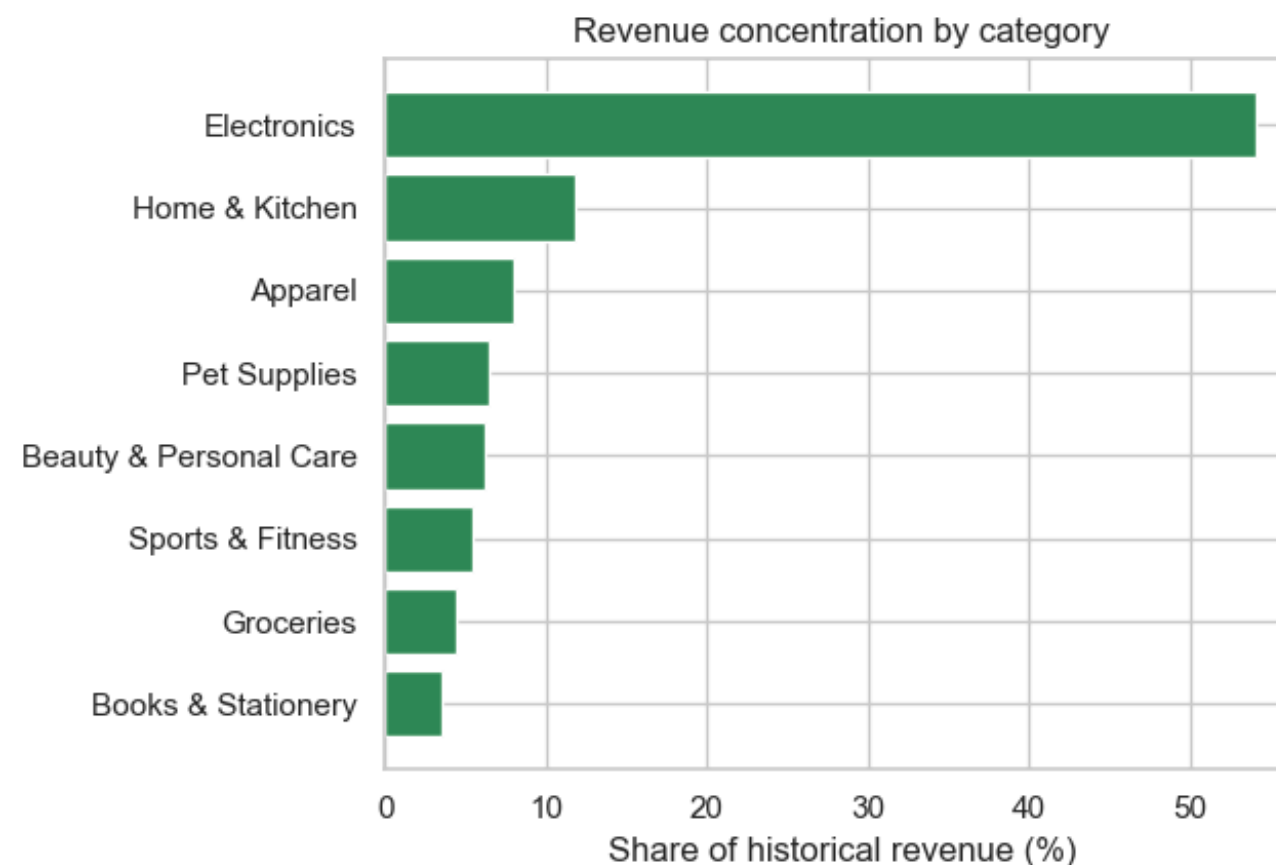
Project Goal & Data overview



- The goal was to analyze 53,704 daily observations from 49 products across 8 categories to provide data-backed pricing recommendations.
- The analysis combined transaction, product cost, competitor price, promotion and inventory data to identify pricing opportunities.
- Python libraries were used to clean and visualize the data, estimate price elasticity and optimize SKU-level prices.
- The recommended portfolio produced an 8.5% modeled gross-profit improvement while limiting the volume decline to 1.3%.

Electronics Dominates Revenue

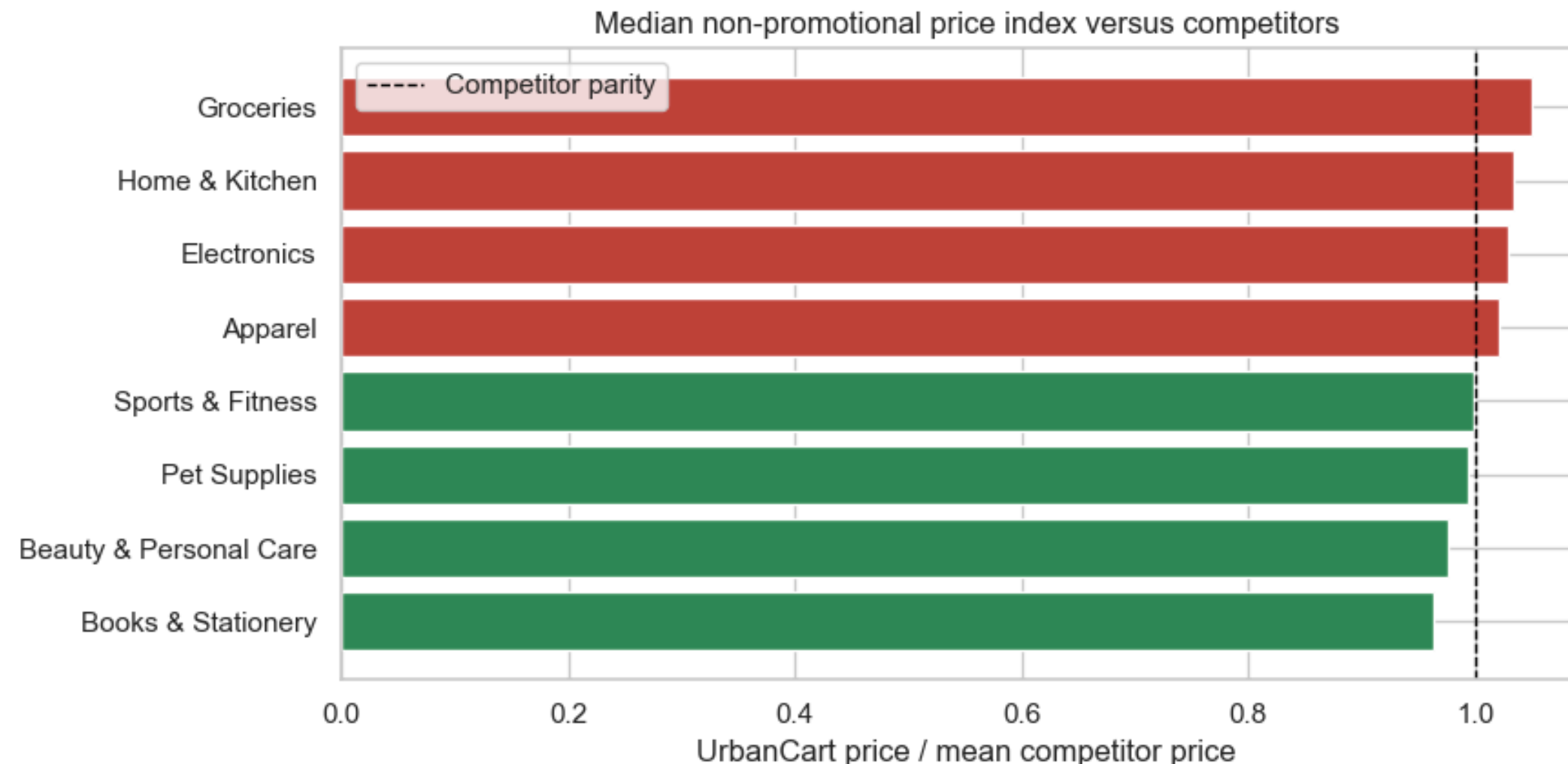
- Electronics contributes more than half of UrbanCart's historical revenue, making it the most commercially important category.
- Home & Kitchen is the second-largest revenue contributor, followed by Apparel and Pet Supplies.
- Most categories operate below their average margin targets, indicating opportunities for selective price improvement.



UrbanCart's Competitive Position



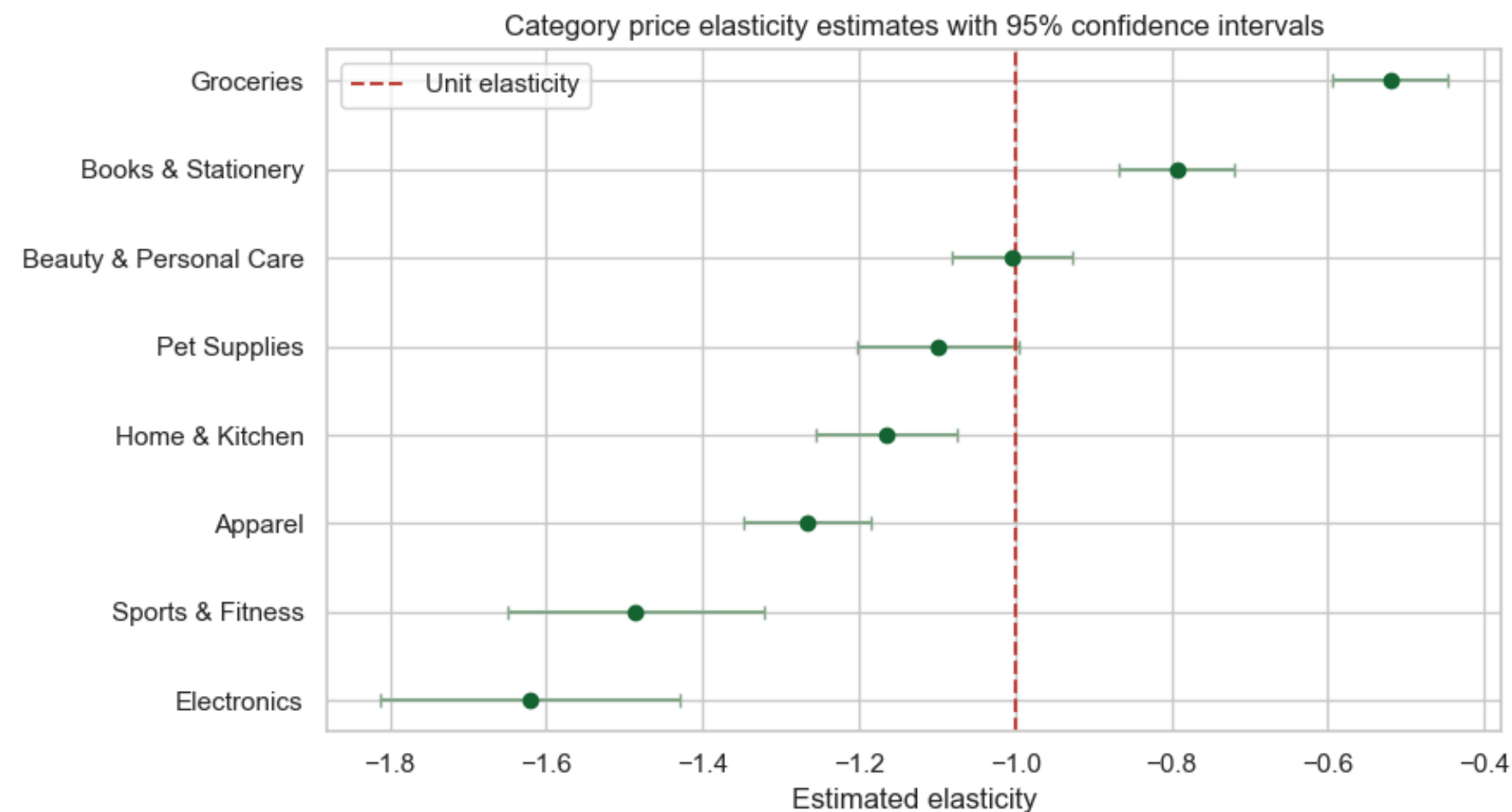
- Groceries, Home & Kitchen, Electronics and Apparel are priced slightly above the average competitor benchmark.
- Beauty & Personal Care and Books & Stationery are priced below competitor parity.
- The variation in market position supports category-specific pricing decisions instead of a uniform pricing rule.



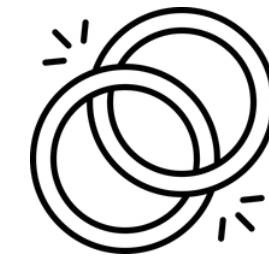
Price Sensitivity Differs Materially



- Electronics and Sports & Fitness are the most price-sensitive categories, with elasticities of -1.62 and -1.49 respectively.
- Groceries and Books & Stationery are relatively less price-sensitive, indicating greater potential pricing flexibility.
- The overall elasticity of -1.16 indicates that demand is generally responsive to changes in price.



Recommended Price Changes



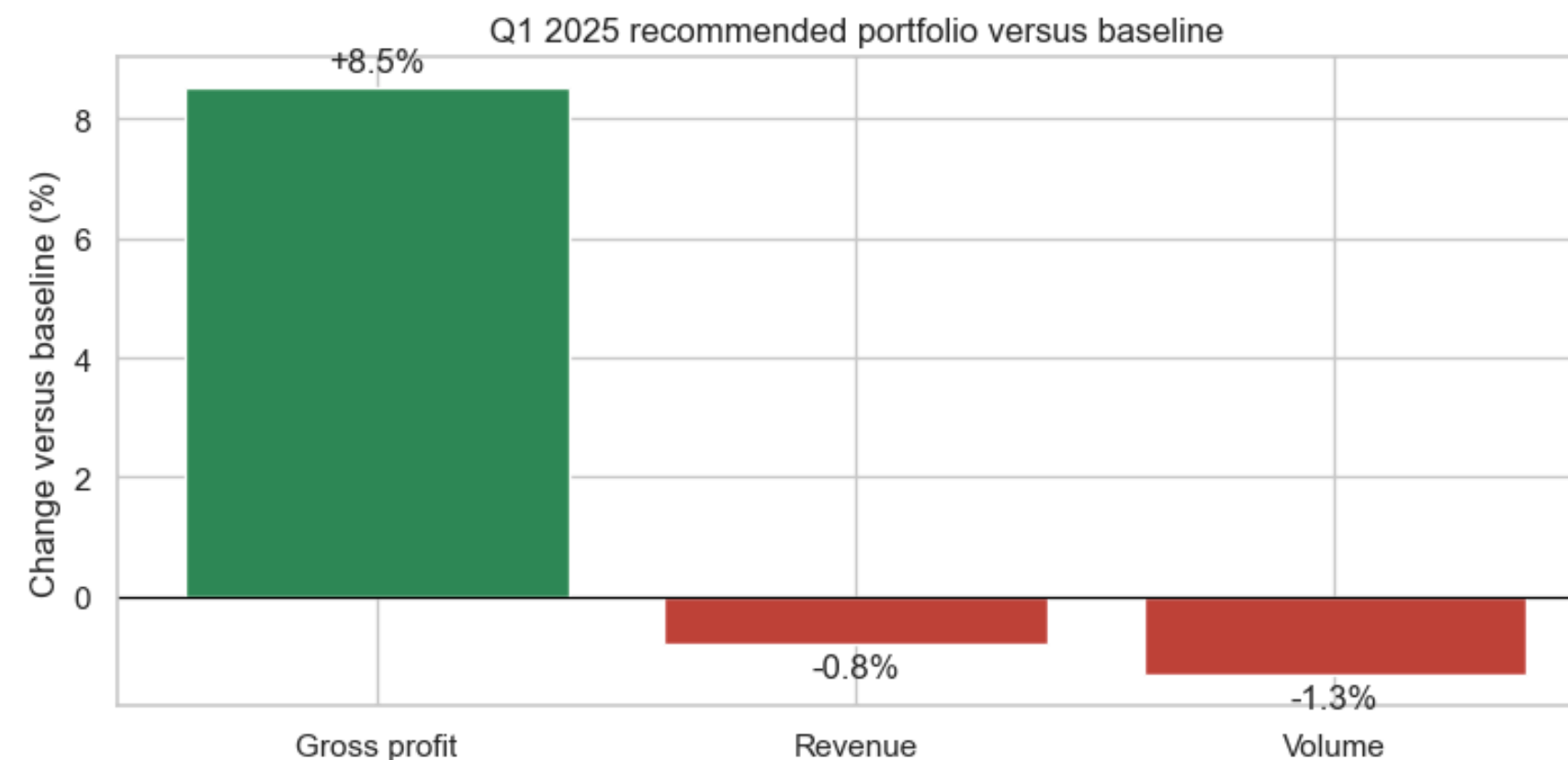
- The model tested price changes between -2.5% and +2.5% for every SKU while considering elasticity, cost and competitor position.
- It selected 24 price increases, 19 price holds and 6 products requiring manual review because their current prices were below cost.
- The largest modelled profit opportunities came from Laptop Air 14, 4K Smart TV, Gaming Console X1 and Vacuum Cleaner.

Product	Current Price	Recommended Price	Change
Laptop Air 14	₹49,424	₹50,659	2.50%
4K Smart TV	₹20,229	₹20,735	2.50%
Gaming Console X1	₹36,484	₹37,397	2.50%
Vacuum Cleaner	₹8,128	₹8,332	2.50%

Recommended Portfolio Meets Both Modeled Targets

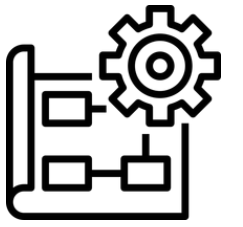


- Gross profit increases from ₹7.19 crore to ₹7.80 crore, representing an improvement of 8.5%.
- Sales volume declines by only 1.3%, remaining comfortably within the permitted 3% limit.
- Gross margin rate improves from 15.6% to 17.1%, while revenue declines marginally by 0.8%.



* Results are modeled forecasts and should be validated through controlled pricing tests.

Implement the Recommendations Through a Controlled Pricing Pilot



- Begin with high-confidence products that offer the largest modeled incremental gross-profit opportunity.
- Use treatment and holdout groups to measure the actual effect of the recommended prices.
- Monitor gross profit, volume, conversion, stockouts and competitor prices every week.
- Automatically roll back a price if the realized volume decline reaches 3%.

Roadmap & Next Steps



Week 1-2:

- Validate current prices, promotions, inventory and competitor information.
- Finalize the high-confidence SKUs to include in the pilot.

Weeks 3-6:

- Launch the controlled pricing pilot with randomized holdout groups.
- Track volume, conversion, revenue and gross profit every week.

Week 7 & Beyond:

- Scale only the prices that deliver positive verified incremental profit.
- Review below-cost products separately with category and procurement teams.
- Retrain the model using experimental results before expanding further.

THANK YOU